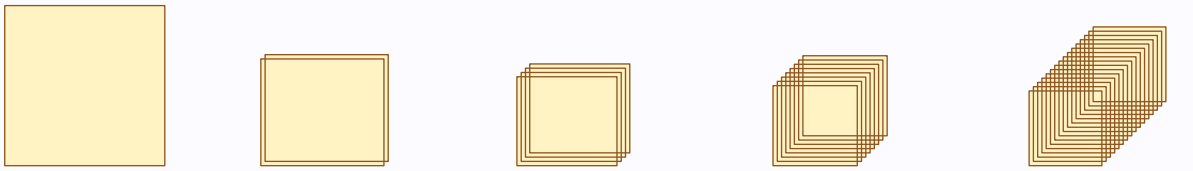
Chapter 9

Indices (powers and roots)

multiplying a number by itself, again and again



$2^0 = 1$ — 0 folds

$2^1 = 2$ — 1 fold

$2^2 = 4$ — 2 folds

$2^3 = 8$ — 3 folds

$2^4 = 16$ — 4 folds

Fold a sheet of paper in half: 2 layers. Again: 4. Again: 8. Each fold **doubles** the layers. After 5 folds you have $2 \times 2 \times 2 \times 2 \times 2 = 32$ layers. Writing all those 2s is tiring, so maths has a shorthand: 2^5. That little raised number is a **power** (also called an **index** or **exponent**).

Powers are everywhere: areas (cm²), volumes (cm³), computer memory (2^{10} bytes), tiny and huge numbers (standard form), and later exponential growth and logarithms (Chapters 21-22). This chapter builds every rule by **counting**, so you never have to memorise blindly.

Goes with **Lecture 7** (power terminology, combining powers, standard form).

A power is repeated multiplication

RULE: what a power means

$a^n = a \times a \times \cdots \times a$ with n copies of a. a is the **base**; n is the **power** (also called the **index** or **exponent**).

$a^1 = a$, a^2 is "a squared", a^3 is "a cubed". a^n is **not** $a \times n$.

Your lecture puts it neatly. **Repeated adding** has a shortcut called multiplying; **repeated multiplying** has a shortcut called a power:

$$3 + 3 + 3 + 3 = 4 \times 3 = 12$$ *four 3s added*

$$3 \times 3 \times 3 \times 3 = 3^4 = 81$$ *four 3s multiplied*

3^4 ← *the power (index, exponent): how many copies of the base*

say: "three to the power four" or "three to the four"

the base: the number being multiplied

Special names: power 2 is "squared" ($5^2 = 25$), power 3 is "cubed" ($2^3 = 8$). Power 1 is just the number itself: $7^1 = 7$ (one copy).

The trap: 3^4 is **not** 3×4. $3^4 = 81$ but $3 \times 4 = 12$. The power counts copies; it doesn't multiply by itself.

Typing powers: calculators have an x^y or ^ key; computers write 3^4, and Python writes 3**4. **Your slides mention "x^2" for when you can't type a raised number.**

$$10^1 = 10 \qquad 10^2 = 100 \qquad 10^3 = 1000 \qquad 10^6 = 1{,}000{,}000$$

Powers of 10 are easy: the power counts the zeros.

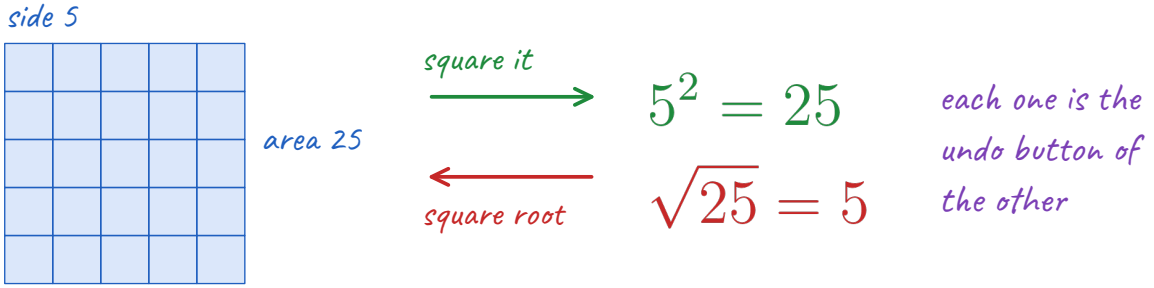
Roots: running a power backwards

RULE: roots

$\sqrt[n]{a}$ (the n-th root of a) is the number whose n-th power is a: $\sqrt[3]{8} = 2$ because $2^3 = 8$. $\sqrt{a}$ means $\sqrt[2]{a}$.

The $\sqrt{\ }$ sign means the **positive** root: $\sqrt{25} = 5$. But $x^2 = 25$ has two solutions, $x = \pm 5$.

A square with side 5 has area $5^2 = 25$. Going backwards, "which side gives area 25?", is the **square root**: $\sqrt{25} = 5$. The √ sign started life as a letter r, for radix (root).



Cube roots undo cubing: a cube of volume 8 has side 2, so $\sqrt[3]{8} = 2$. In general $\sqrt[n]{x}$ (the **n-th root**) undoes the power n: $\sqrt[4]{81} = 3$ because $3^4 = 81$.

Two answers hiding: $(-5)^2$ is also 25 (negative × negative is positive). The √ sign always means the **positive** one, but when solving $x^2 = 25$ (Chapter 15) remember $x = 5$ or $x = -5$, written $x = \pm 5$.

Negative bases, and roots of negative numbers

RULE: powers of a negative number

A negative base to an **even** power is **positive**: $(-2)^4 = 16$. Every pair of minus signs multiplies to $+$.

A negative base to an **odd** power is **negative**: $(-2)^3 = -8$. One minus sign is left over.

Brackets matter: $(-3)^2 = (-3) \times (-3) = 9$, but $-3^2 = -(3 \times 3) = -9$. Without brackets the power is done first and only touches the 3.

RULE: roots of a negative number

Even roots ($\sqrt{}$, $\sqrt[4]{}$, ...) of a negative number **do not exist** as real numbers: no real number squared gives -4, so $\sqrt{-4}$ is undefined (your calculator says Error).

Odd roots of a negative number are negative: $\sqrt[3]{-8} = -2$ because $(-2)^3 = -8$.

$$(-2)^4 = (-2)(-2)(-2)(-2) = 16$$ ← two pairs of minuses: positive

$$(-2)^5 = 16 \times (-2) = -32$$ ← one more minus: negative

$$-2^4 = -(2^4) = -16$$ ← no brackets: the minus waits until the end

$$\sqrt[3]{-27} = -3$$ ← check: $(-3)^3 = -27$

The first law: multiplying powers

RULE: multiplying powers of the same base

$a^m \times a^n = a^{m+n}$: keep the base, **add** the powers.

Only for the **same base**: $x^2 \times y^3$ and $2^3 \times 3^2$ cannot be combined this way.

Only for **multiplying**: $x^2 + x^3$ is not x^5; it can't be simplified.

Numbers in front (coefficients) are multiplied as ordinary numbers: $3x^2 \times 2x^3 = 6x^5$.

Why? What is $x^2 \times x^3$? **Write out every x and count them**:

$$x^2 \times x^3 = \underbrace{x\,x}_{\text{2 x's}} \times \underbrace{x\,x\,x}_{\text{3 x's}} = \underbrace{x^5}_{\text{5 x's altogether}}$$

The powers just **add**. Test with numbers: $2^2 \times 2^3 = 4 \times 8 = 32$ and $2^5 = 32$ ✓.

Example: $3x^2 \times 2x^3$.

$$(3 \times 2)(x^2 \times x^3) = 6x^5 \quad \longleftarrow \text{numbers × numbers; add the powers}$$

Only for the same base. $x^2 \times y^3$ can't be squashed (different envelopes), and $x^2 + x^3$ is **not** x^5 (that's adding, not multiplying: test $x = 2$: $4 + 8 = 12$, but $2^5 = 32$).

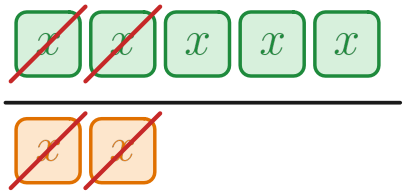
Dividing: cancel in pairs

RULE: dividing powers of the same base

$a^m \div a^n = \dfrac{a^m}{a^n} = a^{m-n}$: keep the base, **subtract** the powers (top minus bottom).

Coefficients are divided as ordinary numbers: $6x^5 \div 2x^2 = 3x^3$.

Why? $\frac{x^5}{x^2}$: write the x's out on top and bottom. Each x on the bottom cancels one on top (like simplifying a fraction, Chapter 2):



$= x^3$

two pairs cancel;
5 − 2 = 3 x's are left

Test: $\frac{2^5}{2^2} = \frac{32}{4} = 8 = 2^3$ ✓.

Example: $6x^5 \div 2x^2$.

$$\frac{6x^5}{2x^2} = \frac{6}{2} \times \frac{x^5}{x^2} = 3x^3$$ ← *divide the numbers; subtract the powers*

Example: $\dfrac{10a^4b^3}{5ab}$.

$$\frac{10}{5}\, a^{4-1}\, b^{3-1} = 2a^3b^2$$ ← *each letter separately; a = a¹*

A power of a power: groups of groups

RULE: powers of powers and of brackets

$(a^m)^n = a^{mn}$: **multiply** the powers.

A power outside a bracket applies to **every factor inside**: $(ab)^n = a^n b^n$ and $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$.

That includes the number in front: $(2x^3)^2 = 2^2 x^6 = 4x^6$, **not** $2x^6$.

Why? $(x^2)^3$ means three copies of x^2. Each copy is a pair of x's, so count the pairs:

$$(x^2)^3 = \boxed{x\,x}\ \boxed{x\,x}\ \boxed{x\,x} = x^6$$

3 groups of 2 = 6 x's: the powers multiply

Brackets with several things inside (the last two laws on your slides). The power applies to **everything** in the bracket:

$$(xy)^3 = xy \times xy \times xy = x^3y^3$$ ← *rearrange: all x's together, all y's together*

$$\left(\frac{x}{y}\right)^3 = \frac{x}{y} \times \frac{x}{y} \times \frac{x}{y} = \frac{x^3}{y^3}$$ ← *top × top, bottom × bottom*

$(2x)^3 = 8x^3$ ← *the 2 gets cubed too! ($2^3 = 8$)*

$(2x^3)^2 = 2^2 \times x^{3\times 2} = 4x^6$ ← *square the 2; multiply the powers*

$(-3a^2b)^3 = (-3)^3 a^6 b^3 = -27a^6b^3$ ← *odd power of −3 stays negative*

The staircase: zero and negative powers

RULE: zero and negative powers

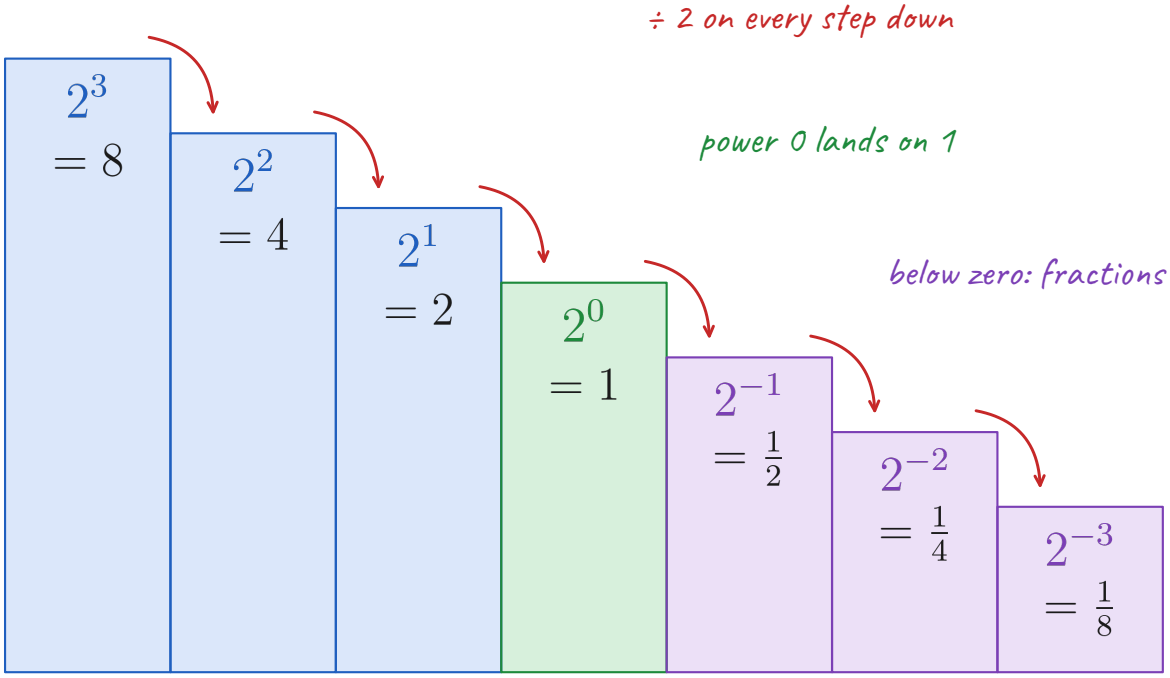
$a^0 = 1$ for any $a \neq 0$.

$a^{-n} = \dfrac{1}{a^n}$: a negative power means **one over** (the reciprocal). It does **not** make the answer negative: $2^{-3} = \frac{1}{8}$.

Turned the other way: $\dfrac{1}{a^{-n}} = a^n$, so $\dfrac{1}{x^{-2}} = x^2$. A power can move between top and bottom of a fraction by changing its sign.

For a fraction, flip it and make the power positive: $\left(\dfrac{a}{b}\right)^{-n} = \left(\dfrac{b}{a}\right)^{n}$.

Why? What could 2^0 or 2^{-1} possibly mean? "Zero copies of 2 multiplied"? Walk down a staircase instead. Every step down **divides by 2** (the opposite of one more fold):



So **anything to the power 0 is 1**, and a **negative power means "one over"** the positive power. "One over" is the **reciprocal**: flip the number upside down. A whole number is secretly over 1, so $8 = \frac{8}{1}$ flips to $\frac{1}{8}$ (Chapter 2, "Dividing"). Read the staircase: 2^{-3} is $2^3 = 8$ flipped, which is $\frac{1}{8}$.

$$\frac{1}{x^{-2}} = \frac{1}{\frac{1}{x^2}} = x^2$$ ← one over a reciprocal flips it back

$$\left(\frac{2}{3}\right)^{-2} = \left(\frac{3}{2}\right)^{2} = \frac{9}{4}$$ ← flip, then square

$$5x^{-3} = \frac{5}{x^3}$$ ← only the x has the power; the 5 stays on top

The dividing law agrees, which is how we know the staircase isn't a trick: $\frac{x^3}{x^3} = x^{3-3} = x^0$, but anything divided by itself is 1. And $\frac{x^2}{x^3} = x^{-1}$, but cancelling leaves $\frac{1}{x}$.

A negative power does NOT make the number negative: $2^{-3} = \frac{1}{8}$, a small positive number.

Fraction powers are roots

RULE: fraction powers

$a^{\frac{1}{n}} = \sqrt[n]{a}$: the **bottom** of the fraction is the **root**.

$a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m = \sqrt[n]{a^m}$: the **top** is an ordinary **power**.

PROCEDURE: evaluating a (negative) fraction power

1. If the power is negative, write one over the same thing with a positive power.
2. Take the root given by the bottom number n (smaller numbers this way round).
3. Raise to the power given by the top number m.

Why? What would $9^{\frac{1}{2}}$ mean? Use the power-of-a-power law as a detective. Square it:

$$\left(9^{\frac{1}{2}}\right)^2 = 9^{\frac{1}{2}\times 2} = 9^1 = 9$$ ← *powers multiply: ½ × 2 = 1*

so $9^{\frac{1}{2}}$ is the number whose square is 9 ← *that's the definition of √*

$$9^{\frac{1}{2}} = \sqrt{9} = 3$$

That is the rule in the box: the bottom of the fraction is the root, the top an ordinary power.

$$8^{\frac{2}{3}} = \left(\sqrt[3]{8}\right)^2 = 2^2 = 4$$ ← *root first (cube root of 8 is 2), then square*

$$16^{-\frac{1}{2}} = \frac{1}{\sqrt{16}} = \frac{1}{4}$$ ← *minus: one over. Half: square root.*

$$27^{-\frac{2}{3}} = \frac{1}{27^{\frac{2}{3}}} = \frac{1}{\left(\sqrt[3]{27}\right)^2} = \frac{1}{3^2} = \frac{1}{9}$$ ← *the procedure: flip, root, power*

Why roots first? Smaller numbers: $(\sqrt[3]{8})^2 = 2^2$ is easier than $\sqrt[3]{64}$ (same answer, 4).

All the index laws in one table

Every law in this chapter, with one example each. Learn the table; the rest of the chapter explains where each line comes from.

Law	Rule	Example
Multiplying	$a^m \times a^n = a^{m+n}$	$x^2 \times x^3 = x^5$
Dividing	$a^m \div a^n = a^{m-n}$	$x^7 \div x^3 = x^4$
Power of a power	$(a^m)^n = a^{mn}$	$(x^2)^3 = x^6$
Zero power	$a^0 = 1 \quad (a \neq 0)$	$5^0 = 1$
Negative power	$a^{-n} = \dfrac{1}{a^n}$	$2^{-3} = \dfrac{1}{8}$
One over a negative power	$\dfrac{1}{a^{-n}} = a^n$	$\dfrac{1}{x^{-2}} = x^2$
Root	$a^{\frac{1}{n}} = \sqrt[n]{a}$	$8^{\frac{1}{3}} = \sqrt[3]{8} = 2$
Fraction power	$a^{\frac{m}{n}} = (\sqrt[n]{a})^m$	$8^{\frac{2}{3}} = 2^2 = 4$
Power of a product	$(ab)^n = a^n b^n$	$(2x^3)^2 = 4x^6$
Power of a fraction	$\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$	$\left(\dfrac{x}{3}\right)^2 = \dfrac{x^2}{9}$
Negative power of a fraction	$\left(\dfrac{a}{b}\right)^{-n} = \left(\dfrac{b}{a}\right)^n$	$\left(\dfrac{2}{3}\right)^{-2} = \dfrac{9}{4}$

RULE: when the laws apply

The first three laws need the **same base**. $2^3 \times 2^4 = 2^7$, but $2^3 \times 3^4$ can only be worked out as numbers (8×81). Never multiply the bases: $2^3 \times 2^4$ is not 4^7.

There are **no laws for adding**: $x^2 + x^3$ and $2^3 + 2^4$ stay as they are (or are worked out as numbers).

Coefficients are ordinary numbers: multiply or divide them separately, but a bracket's power applies to them too.

Find the unknown: equations with a power

Exam questions often give you the answer and hide the power (or the base). Two rules handle almost all of them:

PROCEDURE: unknown power: make the bases match

1. Write both sides as powers of the **same base**: $\frac{1}{9} = 3^{-2}$, $32 = 2^5$, $4 = 2^2$, $8 = 2^3$.
2. If $a^p = a^q$ (same base, not 0, 1 or -1), then the powers are equal: $p = q$.
3. Solve that ordinary equation, then check in the original.

RULE: unknown base: undo the power

If $x^{\frac{m}{n}} = c$, raise both sides to the **flipped** power $\frac{n}{m}$: $x = c^{\frac{n}{m}}$, because $\left(x^{\frac{m}{n}}\right)^{\frac{n}{m}} = x^1$.

Example: solve $3^n = \frac{1}{9}$.

$$\frac{1}{9} = \frac{1}{3^2} = 3^{-2}$$ ← write 1/9 as a power of 3

$$3^n = 3^{-2} \Rightarrow n = -2$$ ← same base: powers equal

Example: solve $2^x = 32$.

$$2^x = 2^5 \Rightarrow x = 5$$ ← 2 × 2 × 2 × 2 × 2 = 32

Example: solve $4^x = 8$. (4 and 8 are both powers of 2.)

$$(2^2)^x = 2^3$$ ← $4 = 2^2$, $8 = 2^3$

$$2^{2x} = 2^3$$ ← power of a power: multiply

$$2x = 3 \Rightarrow x = \frac{3}{2}$$ ← powers equal

Check: $4^{\frac{3}{2}} = (\sqrt{4})^3 = 2^3 = 8$ ✓.

Example: solve $x^{\frac{3}{2}} = 8$.

$$x = 8^{\frac{2}{3}}$$ ← raise both sides to the flipped power 2/3

$$x = \left(\sqrt[3]{8}\right)^2 = 2^2 = 4$$ ← root first, then square

Example: find k if $x^k \times x^3 = x^{10}$.

$$x^{k+3} = x^{10} \Rightarrow k + 3 = 10 \Rightarrow k = 7$$ ← first law, then match powers

Example: solve $5^{x+1} = 125$.

$$5^{x+1} = 5^3 \Rightarrow x + 1 = 3 \Rightarrow x = 2$$ ← $125 = 5^3$

Test before you trust: four famous mistakes

The square-root one is on Lecture 2's "common mistakes" slide; the other three are just as common. Each one is caught instantly by trying small numbers:

$(a+b)^2 \neq a^2 + b^2$ — a = 3, b = 4: $(7)^2 = 49$, but $9 + 16 = 25$

the square is on the whole bracket; Chapter 11 shows $(a+b)^2 = a^2 + 2ab + b^2$

$\sqrt{a^2+b^2} \neq a + b$ — a = 3, b = 4: $\sqrt{25} = 5$, but $3 + 4 = 7$

roots don't split over +

$2x^3 \neq (2x)^3$ — x = 1: $2 \times 1 = 2$, but $2^3 = 8$

the power only sits on what it touches

$x^2 + x^3 \neq x^5$ — x = 2: $4 + 8 = 12$, but 32

laws are for ×, not +

✗ Common mistakes

✗ $2^3 \times 2^4 = 4^7$ ✓ $2^3 \times 2^4 = 2^7$

same base: keep the base, add the powers

✗ $(2x^3)^2 = 2x^6$ ✓ $(2x^3)^2 = 4x^6$

the bracket's power applies to the 2 as well

✗ $3^{-2} = -9$ ✓ $3^{-2} = \frac{1}{9}$

a negative power means one over, not a negative answer

✗ $-3^2 = 9$ ✓ $-3^2 = -9, \quad (-3)^2 = 9$

without brackets the power touches only the 3

✗ $x^6 \div x^2 = x^3$ ✓ $x^6 \div x^2 = x^4$

dividing subtracts the powers; it doesn't divide them

✗ $x^0 = 0$ ✓ $x^0 = 1$

anything (except 0) to the power 0 is 1

Standard form: KB, MB, GB and tiny things

RULE: standard form

A number in standard form is $a \times 10^n$ with $1 \leq a < 10$ and n a whole number (positive, negative or zero).

Numbers ≥ 10 have a positive power; numbers between 0 and 1 have a negative power; numbers from 1 to 9.99... have power 0.

A 1 GB file holds about 1,000,000,000 bytes. A bacterium is 0.000002 m long. Counting zeros is error-prone, so we write a **small number × a power of 10**:

1 kB	1000 bytes	1×10^3
1 MB	1,000,000 bytes	1×10^6
1 GB	1,000,000,000 bytes	1×10^9
57		5.7×10^1
0.0024		2.4×10^{-3}
0.000001		1×10^{-6}

big numbers: positive power

small numbers: negative power

The recipe. Write the digits with the dot after the first non-zero digit (that's a, between 1 and 10). Then count how many places the digits had to slide (Chapter 3): that's the power. **Digits slid left** (the number was small, less than 1) → **negative** power. Digits slid right (the number was big) → positive power.

$0.0024 \rightarrow 2.4$ ← *digits slide 3 places left, so the dot seems to move right*

$0.0024 = 2.4 \times 10^{-3}$ ← *× 10^{-3} (÷ 1000) puts it back*

Slide note: Lecture 7 writes $1 \leq a \leq 10$; it should be $1 \leq a < 10$. The number 10 itself is written 1×10^1, never 10×10^0.

Multiplying in standard form uses the first law on the powers of 10: $(3 \times 10^4) \times (2 \times 10^5) = 6 \times 10^9$. Calculators show 2.4×10^{-3} as 2.4E-3 (E for exponent).

Calculating in standard form

PROCEDURE: multiplying and dividing

1. Multiply (or divide) the front numbers.
2. Add the powers of 10 when multiplying; subtract them when dividing.
3. **Re-normalise**: if the front number is 10 or more, divide it by 10 and add 1 to the power ($20 \times 10^5 = 2 \times 10^6$); if it is less than 1, multiply it by 10 and take 1 off the power ($0.5 \times 10^3 = 5 \times 10^2$).

PROCEDURE: adding and subtracting

1. Rewrite the numbers with the **same power of 10** first (or write them out in full).
2. Add or subtract the front numbers; keep the power.
3. Re-normalise if needed.

Example: $(6 \times 10^5) \times (5 \times 10^3)$.

$$(6 \times 5) \times 10^{5+3} = 30 \times 10^8$$ ← multiply fronts, add powers

$$= 3 \times 10^9$$ ← 30 is too big: 30 = 3 × 10^1, so add 1 to the power

Example: $(8 \times 10^7) \div (2 \times 10^3)$.

$$\frac{8}{2} \times 10^{7-3} = 4 \times 10^4$$ ← divide fronts, subtract powers

Example: $(3 \times 10^4) \div (6 \times 10^{-2})$.

$$\frac{3}{6} \times 10^{4-(-2)} = 0.5 \times 10^6$$ ← 4 − (−2) = 6

$$= 5 \times 10^5$$ ← 0.5 is less than 1: × 10 and take 1 off the power

Example: $(3 \times 10^4) + (5 \times 10^3)$.

$$3 \times 10^4 + 0.5 \times 10^4$$ ← $5 \times 10^3 = 0.5 \times 10^4$: same power now

$$= 3.5 \times 10^4$$ ← add the fronts, keep the power

Check by writing them out: $30\,000 + 5000 = 35\,000$ ✓. **Never** add the powers when adding: $3 \times 10^4 + 5 \times 10^3$ is not 8×10^7.

Example: $(7.2 \times 10^{-3}) - (4 \times 10^{-4})$.

$$7.2 \times 10^{-3} - 0.4 \times 10^{-3} = 6.8 \times 10^{-3}$$ ← $4 \times 10^{-4} = 0.4 \times 10^{-3}$

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Power	$a^n = a \times a \times \cdots \times a$ n copies of a; $a^1 = a$
Multiplying	$a^m \times a^n = a^{m+n}$ same base only; coefficients multiply
Dividing	$\dfrac{a^m}{a^n} = a^{m-n}$ same base only; coefficients divide
Power of a power	$(a^m)^n = a^{mn}$
Brackets	$(ab)^n = a^n b^n, \quad \left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$ the number in front is raised too: $(2x^3)^2 = 4x^6$
Zero power	$a^0 = 1 \quad (a \neq 0)$
Negative power	$a^{-n} = \dfrac{1}{a^n}, \quad \dfrac{1}{a^{-n}} = a^n$ one over; never a negative answer
Negative power of a fraction	$\left(\dfrac{a}{b}\right)^{-n} = \left(\dfrac{b}{a}\right)^n$
Fraction power	$a^{\frac{1}{n}} = \sqrt[n]{a}, \quad a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m$ bottom = root, top = power
Negative base	$(-2)^4 = 16, \ (-2)^3 = -8, \ -3^2 = -9$ even power positive, odd power negative
Roots of negatives	$\sqrt[3]{-8} = -2$ odd roots fine; even roots don't exist (real numbers)
No adding law	$x^2 + x^3$ and $(a+b)^2$ don't follow the index laws
Unknown power	$a^p = a^q \Rightarrow p = q$ write both sides as powers of the same base
Unknown base	$x^{\frac{m}{n}} = c \Rightarrow x = c^{\frac{n}{m}}$
Standard form	$a \times 10^n, \quad 1 \leq a < 10$ × / ÷: fronts, then powers, then re-normalise
Adding in standard form	make the powers of 10 equal first, then add the fronts

Folding paper: your turn

Answers in grey. Write out the letters longhand if you're not sure!

Values

1. 2^6
2. 5^3
3. 10^0
4. 4^{-1}
5. 3^{-2}
6. 25^(1/2)
7. 27^(2/3)
8. √(49)

1. 64 2. 125 3. 1 4. 1/4 5. 1/9 6. 5 7. 9 8. 7

Simplify

9. $x^4 \times x^5$
10. $y^9 \div y^3$
11. $(a^3)^4$
12. $(3x)^2$
13. $x^{-2} \times x^5$
14. $(x^2y^3)^2$

9. x^9 10. y^6 11. a^{12} 12. $9x^2$ 13. x^3 14. x^4y^6

Standard form

15. Write 45,000 in standard form
16. Write 0.00061 in standard form
17. $(4 \times 10^3) \times (2 \times 10^{-5})$

15. 4.5×10^4 16. 6.1×10^{-4} 17. 8×10^{-2}

Spot the error

18. Tom says $3^{-2} = -9$. Why is he wrong?

18. negative power means one over: 1/9

Coefficients and negatives

19. $6x^5 \div 2x^2$
20. $(2x^3)^2$
21. $12a^6b^2 \div 4a^2b$
22. $(-2)^5$
23. -2^4
24. $\sqrt[3]{(-27)}$
25. $1/x^{-3}$
26. $(3/4)^{-2}$

19. $3x^3$ 20. $4x^6$ 21. $3a^4b$ 22. -32 23. -16 24. -3 25. x^3 26. $16/9$

Calculating in standard form

27. $(6 \times 10^5) \times (5 \times 10^3)$
28. $(9 \times 10^8) \div (3 \times 10^{-2})$
29. $(4 \times 10^3) + (2 \times 10^2)$
30. Write 25×10^4 in standard form

27. 3×10^9 28. 3×10^{10} 29. 4.2×10^3 30. 2.5×10^5

Find the unknown

31. $3^n = 1/9$
32. $2^x = 32$
33. $4^x = 8$
34. x^(3/2) = 8
35. $x^k \times x^3 = x^{10}$
36. 5^(x+1) = 125

31. $n = -2$ ($1/9 = 3^{-2}$) 32. $x = 5$ ($32 = 2^5$) 33. $x = 3/2$ ($2^{2x} = 2^3$) 34. x = 4 (x = 8^(2/3))
35. $k = 7$ ($k + 3 = 10$) 36. $x = 2$ ($125 = 5^3$, $x + 1 = 3$)